

Assignment IV

(Advance Vector Calculus)

Subject: Calculus-II (MTH 210)

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Tutor: Ramchandra Timilsina

ramchandra.timilsina@eemc.edu.np

(Vector & Scalar point function:)

1. Find the velocity and acceleration of the moving particle along the curve $\vec{r} = (t, t^2, t^3)$ at $t = 1$. **Ans:**

2. Find the angle between the curve $\vec{r} = (t^2, 3t^2, -(t^2 + 1))$ at the point $t = \pm 1$. **Ans:**

3. A particle moves along the curve $x = t^3 + 1, y = t^2, z = 2t + 5$. Find the component of velocity and acceleration at $t = 1$ along $\vec{i} + \vec{j} + 3\vec{k}$. **Ans:** $\sqrt{11}, \frac{8}{\sqrt{11}}$

4. If $\vec{r} = (3t, 3t^2, 2t^3)$ then show that $[\dot{\vec{r}} \ddot{\vec{r}} \dddot{\vec{r}}] = 216$.

Solution: Example-5, page: 172.

5. Find the directional derivative of $f = x^2 - y^2 + 2z^2$ at $(1, 2, 3)$ along $\vec{a} = \vec{i} + \vec{j} + \vec{k}$. **Ans:** $\frac{10}{\sqrt{3}}$

6. Find the directional derivative of $f = x^2 - y^2 + 2z^2$ at $P(1, 2, 3)$ along PQ where Q is the point $(5, 0, 4)$. **Ans:**

Hints: First find the equation of line in vector form using $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$. Find its unit vector $\hat{PQ} = \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|}$ then the required directional derivative can be obtained by using $D_{\hat{PQ}}(f) = \nabla f \cdot \hat{PQ}$.

7. Find the directional derivative of $f = 4xz^3 - 3x^2yz^2$ at $(2, -1, 2)$ along z -axis (i.e $\vec{a} = \vec{k}$). **Ans:** 144

8. If $\phi = \log(x^2 + y^2 + z^2)$ then find $\text{div}(\text{grad } \phi)$ and $\text{curl}(\text{grad } \phi)$.

Ans: $\text{div}(\text{grad } \phi) = \frac{2}{x^2 + y^2 + z^2}$ and $\text{curl}(\text{grad } \phi) = 0$

9. Show that $\vec{F} = (y \sin z - \sin x)\vec{i} + (x \sin z + 2yz)\vec{j} + (xy \cos z + y^2)\vec{k}$ is irrotational (conservative) and then find the scalar potential function ϕ such that $\vec{F} = \nabla \phi$.

Ans: $\phi = 3xy \sin z + \cos x + 2y^2z + c$

Solution: Example:32, page-208.

10. Show that the value under the integral sign (i.e integrand) is exact in the domain and evaluate the integral:

(i) $\int_{(1,0,2)}^{(-2,1,3)} [(6xy^3 + 2z^2)dx + 9x^2y^2dy + (4xz + 1)dz]$ **Ans:** -31

Solution: Example:36, page:212.

$$(ii) \int_{(0,0,0)}^{(4,1,2)} [3ydx + 3xdy + 2zdz]$$

Ans:

(Line Integral:)

11. Evaluate $\oint_C [(x^3 - 3y)dx + (x + \sin y)dy]$, where C is the boundary of the triangle with vertices $(0, 0), (1, 0), (0, 2)$. **Ans:** 4

Solution: Example:38, page:217.

12. Verify Green's theorem for $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ over a region R on xy - plane bounded by lines $x = 0, y = 0, x = a, y = b$.

Solution: Example:40, page:219.

13. Using Green's theorem, Evaluate the following line integrals:

(i) $\oint_C [\sqrt{y}dx + \sqrt{x}dy]$, where C : triangle with vertices $(1,1), (3,1)$ and $(2,2)$. **Ans:** $2 - \frac{3}{\sqrt{2}}$

(ii) $\oint_C [5xydx + x^3dy]$, where C : region bounded the graph of $y = x^2$ and $y = 2x$ between the points $(0,0)$ and $(2,4)$. **Ans:** 28/15

(iii) $\oint_C [(3x^2 + y)dx + 4y^2dy]$, where C : boundary of the triangle $(0,0), (1,0), (0,2)$. **Ans:** -1

(iv) $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (x^2e^y, y^2e^x)$ and C : rectangle with vertices $(0,0), (2,0), (2,3)$ and $(0,3)$. **Ans:** $9(e^2 - 1) - \frac{8}{3}(e^3 - 1)$

(v) $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (\frac{e^y}{x}, e^y \ln x + 2x)$ and C : region given by $1 + x^4 \leq y \leq 2$. **Ans:** 16/5

14. Evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$ using Stoke's theorem:

(i) $\vec{F} = (y^2, 2xy + \sin x, 0)$ and C : $0 \leq x \leq \pi/2, 0 \leq y \leq 2$. **Ans:** 2

(ii) $\vec{F} = -5y\vec{i} + 4x\vec{j} + z\vec{k}$ and C : $x^2 + y^2 = 4, z = 1$. **Ans:** 36π

(iii) $\vec{F} = (1, xy, yz)$ and S : $x^2 + y^2 \leq z, y \geq 0, z \leq 4$. **Ans:**

(iv) $\vec{F} = (y, \frac{z}{2}, \frac{3y}{2})$ and C : $x^2 + y^2 + z^2 = 6z, z = x + 3$. **Ans:** $-9\sqrt{2}\pi$

(v) $\vec{F} = (y, xz^3, -zy^3)$ and C : $x^2 + y^2 = 4, z = 3$. **Ans:** -112π

Solution: Example:58, page-242.

(vi) $\vec{F} = (2x - y - z)\vec{i} + (x + y + z^2)\vec{j} + (3x - 2y + 4z)\vec{k}$ and C : $x^2 + y^2 = a^2, z = 0$. **Ans:**

(vii) $\vec{F} = (4z, -2x, 2x)$ and C : $x^2 + y^2 = 1, z = y + 1$. **Ans:** -4π

(viii) $\vec{F} = (y, z, x)$ and C : the surface of sphere $x^2 + y^2 + z^2 = 1$ above the xy - plane. **Ans:**

(ix) $\vec{F} = (y^3, 0, x^3)$ and C : boundary of the triangle with vertices $(1,0,0), (0,1,0), (0,0,1)$. **Ans:** -1/2

(x) $\vec{F} = -3y\vec{i} + 3x\vec{j} + z\vec{k}$ and S: along circle $x^2 + y^2 = 4, z = 1$ **Ans:** 24π

15. Use Stoke's theorem to evaluate $\oint_C [4ydx + 2zdy + 6ydz]$ where C is the curve of intersection of $x^2 + y^2 + z^2 = 6z$ and $z = x + 3$. **Ans:** $36\sqrt{2}\pi$

Solution: Example:57, page-241.

16. Find $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (e^{x+y}, e^{x-y})$ and C: boundary of the triangle $x \leq y \leq 2x, 0 \leq x \leq 1$ by using

(i) Green's theorem (ii) Stoke's theorem

Ans: $\frac{e^2}{2} - \frac{e^3}{3} + \frac{1}{e} - \frac{1}{6}$

(Surface Integral:)

17. Evaluate $\iint_S \vec{F} \cdot \vec{n} ds$ where,

(i) $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ and $\vec{r} = (u \cos v, u \sin v, 3v)$; $0 \leq u \leq 1, 0 \leq v \leq 2\pi$. **Ans:** $12\pi^2$

(ii) $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ and S: the portion of the plane $2x + y + z = 6$ in the first octant.

Ans: 18

(iii) $\vec{F} = yz\vec{i} + zx\vec{j} + xy\vec{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ in the first octant.

Ans: $\frac{3}{8}$

Solution: Example:52, page-233.

(iv) $\vec{F} = (x^2, 0, 3y^2)$ and S: portion of the plane $x + y + z = 1$ in the first octant. **Ans:** $1/2$

(v) $\vec{F} = (3x, 3y, z)$ and S: part of the surface $z = 9 - x^2 - y^2$ with $z \geq 0$. **Ans:** $\frac{567\pi}{2}$

18. Find $\iint_S \vec{F} \cdot \vec{n} ds$ using Gauss divergence theorem:

(i) $\vec{F} = y^2 e^z \vec{i} + xy\vec{j} + x \tan^{-1} y \vec{k}$ and S: $x + y + z = 1$ in the first octant. **Ans:** $-\frac{1}{24}$

(ii) $\vec{F} = (4x, x^2 y, -x^2 z)$ where S is the surface of the tetrahedron with vertices (0,0,0), (1,0,0), (0,1,0) and (0,0,1).

Ans: $\frac{2}{3}$

Solution: Example:67, page-255.

(iii) $\vec{F} = (x, 2y, 4z)$ and S: the surface of the cone $\sqrt{x^2 + y^2} \leq z, 0 \leq z \leq 2$. **Ans:**

(iv) $\vec{F} = (2x^2, \frac{y^2}{2}, -\cos \pi z)$ and S: tetrahedron with vertices (0,0,0), (1,0,0), (0,1,0) and (0,0,1). **Ans:**

(v) $\vec{F} = (2x^2 y, -y^2, 4xz^2)$ and S: $y^2 + z^2 = 3, 0 \leq x \leq 2, y \geq 0, z \geq 0$. **Ans:**

19. Evaluate $\iint_S \vec{F} \cdot \vec{n} dA$ where $\vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ and S is the surface of the region bounded by $x^2 + y^2 = 4, z = 0, z = 3$. **Ans:** 84π

(i) By using direct method

Solution: Same as Examples: 48,49, page-229.

(ii) By using Gauss divergence theorem

Solution: Example:65, page-254.